

RAHUL DEBNATH

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Professional Summary

AI Developer Intern at RMSI with end-to-end experience building and deploying deep learning pipelines for international geospatial projects. Specializes in CNN-based image classification, SAR/optical satellite data analysis, and LLM-powered applications. **IEEE IGARSS 2026** accepted paper on ML-based mining detection. M.Sc. in Data Analytics with Geoinformatics.

Professional Experience

RMSI Pvt. Ltd.

Jan 2026 – Present

AI/ML Intern

Noida, Uttar Pradesh, India

- Delivered an **end-to-end, production-ready deep learning pipeline** for an international client project (Calgary, Canada) — spanning data cleaning, data engineering, feature engineering, model development, training, evaluation, and FastAPI deployment.
- Architected and trained a **ResNet + ImageNet-based CNN model** for geospatial image classification, achieving deployment-ready performance on real-world satellite and aerial imagery.
- Deployed ResNet-based image classifier as a FastAPI microservice, reducing manual geospatial analysis time for an international client project (Calgary, Canada); model achieved deployment-ready accuracy on real-world aerial imagery.
- Actively contributing to **4–5 concurrent geospatial AI, DSS pipeline, and ML/DL model development & deployment projects** across diverse regions including Afghanistan, Jammu & Kashmir, Odisha, and the Konkan–Mumbai corridor — covering applications in terrain analysis, land use mapping, environmental monitoring, predictive analytics, and decision-support systems.

Adamas University

Jun 2025 – Jul 2025

GIS Analyst Intern

Onsite, Kolkata, India

- Executed data preprocessing, feature engineering, and statistical analysis on large geospatial datasets for the research project "*Quantifying the Influence of Adjacent Climate Zones on Land Surface Temperature*".
- Built regression and ML models on 10,000+ spatial data points to quantify climate zone influence on Land Surface Temperature, contributing findings to a peer-reviewed publication in progress.
- Leveraged Python-based data science libraries (**NumPy, Pandas, Scikit-learn**) alongside ArcGIS and QGIS to process, analyze, and visualize spatial datasets; co-authored analytical research documentation under academic supervision.

Skills

- **Programming & Databases:** Python, SQL, PostgreSQL, [NumPy, Pandas, Scikit-learn]
- **AI & Machine Learning:** Generative AI, Large Language Models (LLMs), RAG, Transformers (Encoder–Decoder), KNN, SVM, Naïve Bayes, Decision Trees, Random Forest, Linear/Logistic Regression
- **Deep Learning & Frameworks:** CNN (ResNet, ImageNet), ANN, RNN, TensorFlow, PyTorch
- **Computer Vision:** Image Classification, Object Detection, Image Segmentation, Multimodal AI, Vision–Language Models
- **Natural Language Processing (NLP):** Text Mining, Tokenization, Named Entity Recognition (NER), Sentiment Analysis, Information Retrieval (Boolean, Vector, Probabilistic)
- **API Development & Deployment:** FastAPI, Gradio, Groq API, Production Model Deployment
- **Geospatial Technologies:** GIS, Remote Sensing (RS), Spatial Data Analytics, Change Detection, Image Classification, Geostatistics, SAR Analysis
- **Geospatial Tools & Libraries:** QGIS, ArcGIS, SAGA GIS, Google Earth Engine (GEE), SNAP, GDAL, Rasterio, GeoPandas, Shapely
- **Statistics & Predictive Analytics:** Hypothesis Testing, Regression Analysis, Time Series Forecasting, Feature Engineering
- **Big Data Technologies:** Apache Sedona (GIS Spatial Analytics), Hadoop, PySpark
- **Data Visualization & BI:** Tableau, Matplotlib, Seaborn, Microsoft Excel
- **Cloud & DevOps:** Google Cloud Platform (GCP), Git, GitHub
- **Other Tools:** Jupyter Notebook, Streamlit, LaTeX, Canva, Power Point

Education

M.Sc. Data Analytics with Specialization in Geoinformatics

Aug 2024 – Present

Kerala University of Digital Sciences, Innovation and Technology (Formerly IIITM-K)

Thiruvananthapuram, Kerala, India

Bachelor of Science in Geography (Honours)

2021 – 2024

Ranaghat College, Affiliated with Kalyani University

Ranaghat, Kalyani, West Bengal, India

Projects

PolicyMind-AI — Agentic RAG Insurance Document Intelligence

2026

- * *Production-deployed product at RMSI Private Limited* for extracting structured insurance data from heterogeneous policy PDFs into a standardized JSON schema.
- * Engineered an end-to-end **Agentic AI and RAG pipeline** combining OCR, page classification, deterministic extraction, semantic retrieval, specialist LLM agents, evidence validation, confidence scoring, and human-correction memory.
- * Developed production **FastAPI** services and an evidence-grounded PDF chatbot using **Gemini, Groq, FAISS/pgvector, Hugging Face embeddings, Python, and PostgreSQL**.
- * **Skills:** Agentic AI, RAG, LLMs, Prompt Engineering, OCR, NLP, Vector Search, FastAPI, FAISS, PostgreSQL, Python

Urban Tree Species Classification System: Geospatial AI Product for Calgary

2026

- * *Internship Project, RMSI Private Limited* City of Calgary, Canada Government Project **Production Product**
- * Developed an end-to-end **geospatial AI/ML pipeline** to classify **9 urban tree species** from high-resolution **0.1 m RGB aerial orthomosaic imagery**, supporting automated urban tree inventory management at city scale.
- * Trained and evaluated deep learning models using **PyTorch, Torchvision, and ResNet18 ImageNet transfer learning**, achieving **95.15% validation accuracy** and outperforming a custom CNN baseline of **54.21%**.
- * Engineered large-scale remote sensing workflows for **490 GB GeoTIFF imagery**, including **MrSID-to-GeoTIFF conversion**, image patch extraction, class-balanced dataset creation, geospatial quality control, and VRT-based raster optimization.
- * Deployed the trained model as a **FastAPI REST service** with database-driven inference, background job tracking, error logging, and integration readiness for enterprise AI/ML APIs.
- * **Skills:** Machine Learning, Deep Learning, Computer Vision, Remote Sensing, Geospatial AI, Transfer Learning, PyTorch, Torchvision, ResNet18, FastAPI, Rasterio, GeoPandas, GDAL, QGIS, OpenCV, NumPy, scikit-learn, GeoTIFF, VRT, MLOps

Multi-Temporal Detection of Illegal Iron Ore Mining — Jharkhand & Odisha

Nov 2025

- * *Mini Project (3rd Semester), Digital University Kerala*
- * **[IGARSS 2026 ACCEPTED — Manuscript #2043]** Research paper titled "*Multi-Temporal Detection of Small-Scale Iron Ore Mining Areas Using Integrated SAR and Optical Data with Machine Learning*" **accepted for oral/poster presentation at IEEE IGARSS 2026** by the Technical Program Committee — one of the most prestigious international remote sensing conferences.
- * Developed a machine learning-based change detection workflow using multi-temporal **Sentinel-1 (SAR)** and **Sentinel-2 (optical)** data, integrated with spectral indices (NDVI, NDBI, BSI) in **Google Earth Engine** and **QGIS** for illegal mining area identification and spatial visualization.
- * **Skills:** Remote Sensing, ML, SAR, Sentinel-1/2, GEE, Python, QGIS

Med-Echo: Multimodal AI-Powered Medical Voice Assistant

2025

- * *Generative AI Course Project, Digital University Kerala* [in LinkedIn](#) — [GitHub](#)
- * Developed an **AI-powered medical consultation system** integrating **Whisper V3** (speech recognition), **Llama 4 Scout** (vision-language model), and **neural TTS** for real-time multimodal clinical query interpretation.
- * Built a **Gradio web interface** for voice input and image upload, achieving **<15s end-to-end latency**, **93% accuracy** in medical image interpretation, and **<10% Word Error Rate** in speech transcription.
- * **Skills:** LLMs, VLMs, Speech Recognition, TTS, Python, Groq API, Gradio, Multimodal AI

Generative AI for Synthetic Soil Nutrient Map Generation

2025

- * *Spatial Data Analytics Course Project, Digital University Kerala* [GitHub](#)
- * Designed a **Diffusion Model-based Generative AI system** to simulate and impute missing soil nutrient data, integrating **remote sensing datasets** and **SoilGrids** for enhanced spatial prediction in sustainable agriculture.
- * **Skills:** Generative AI, Diffusion Models, GeoAI, Python, ML, Remote Sensing

Certifications

- * Generative AI with Large Language Models — *DeepLearning.AI & Amazon Web Services (AWS)*
- * Google Cloud Data Analytics Learning Path — *Google Cloud*
- * Google Cloud Computing Foundations Certificate — *Google Cloud*

Additional Information

- * **Languages:** English (Professional), Hindi (Fluent), Bengali (Native)
- * **Research Interests:** AI/ML, GeoAI, Satellite Image Analysis, Multimodal AI, Environmental Remote Sensing, Deep Learning for Earth Observation